

TIME ECHO

Game Design Document

Version 1.0 | Complete

Engine: Unreal Engine | Language: C++

Genre: Puzzle-Platformer | Perspective: Third-Person

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1. Game Overview

Time Echo is a first-person puzzle-platformer developed in Unreal Engine using C++. The game centres on a single innovative mechanic: the Time Echo system. Players record their own actions for a short period, then replay those actions as a temporal clone. By cooperating with their past self, players solve environmental puzzles that would be impossible to complete alone.

The project is at a complete and polishing stage, featuring 20 handcrafted levels that progressively introduce and combine every core mechanic.

Title	Time Echo
Genre	First-Person Puzzle-Platformer
Engine	Unreal Engine (C++)
Development Stage	Complete / Polishing
Total Levels	20
Core Mechanic	Time Echo Recording & Clone Replay System
Player Count	Single Player

2. Core Mechanic: The Time Echo System

The Time Echo system is the foundation of every puzzle in the game. It allows players to record their movement and interactions, then replay that sequence as an autonomous clone that acts alongside them in real time.

2.1 Recording Phase

When the player activates recording (Q), the system begins capturing their state at fixed time intervals, independent of frame rate. Each recorded snapshot — called a frame — stores the following data:

- World position (FVector Location)
- Character rotation (FRotator Rotation)
- Timestamp in seconds since recording started
- Whether the player was sprinting (bIsSprinting)
- Whether the player was jumping or airborne (bIsJumping)
- Whether the player was triggering an interact action (bIsInteracting)

Recording lasts for a fixed maximum duration (approximately 10 seconds). The player can end recording early. All frames are stored in an ordered array inside the TERERecordingComponent.

2.2 Clone Replay Phase

When the player activates replay (R), a clone actor (ATECloneActor) is spawned in the world. The recorded frame data is passed to the clone via `InitialiseWithFrames()`. The clone then:

- Begins at the position and rotation of the first recorded frame
- Advances through frames in real time using linear interpolation between positions
- Smoothly interpolates yaw rotation between frames
- Replicates sprinting and jumping states for animation fidelity
- Fires interaction events when `bIsInteracting` was true, triggering nearby buttons within radius
- Freezes in place at the last recorded frame if the recording ended before the fixed lifetime
- Remains in the world for the full duration of `TotalLifetime` before disappearing

2.3 Clone Lifetime & Freezing

If the recording is shorter than the fixed time limit, the clone plays back all recorded frames and then freezes at its final position. This is a key design feature: a frozen clone can hold a pressure plate or block a laser indefinitely until its lifetime expires, giving the player time to act.

Three delegate events are broadcast by the clone throughout its lifecycle:

- OnCloneSpawned: fires immediately when the clone is initialised
- OnCloneInteracted: fires each time the clone activates a nearby button
- OnCloneDestroyed: fires just before the clone is removed from the world

3. Player System

3.1 Controls

W / A / S / D	Movement (forward, left, back, right)
Mouse	Look around (first-person camera)
Shift	Sprint
Space	Jump
E	Interact with buttons or objects
Q	Start / Stop recording
R	Replay recording — spawns the clone

3.2 Quality-of-Life Features

The player has two reset options available at any time to prevent them from becoming stuck:

- **Reset to Checkpoint:** teleports the player back to the most recently reached checkpoint. Active recordings are stopped and the current clone is destroyed.
- **Restart Level:** teleports the player all the way back to the level start checkpoint, resetting progression within the level.

Both actions are managed by `TELevelManager`, which automatically discovers and sorts all checkpoint actors in the level at `BeginPlay`. Checkpoints only advance forward; reaching an earlier checkpoint after a later one has no effect.

4. Environmental Systems

Time Echo features a set of purpose-built environmental actors that form the vocabulary of its puzzles. Each system is designed to be activatable by both the player and the clone.

4.1 Buttons (ATEButton)

Buttons are discrete interactable objects. They can be activated by the player pressing E within range, or by the clone when `bIsInteracting` was recorded near the button. Activation calls through the `TEInteractableBase` interface, which notifies any linked actors (typically doors) to open or close. Buttons can be configured to allow multiple activations or only a single one.

4.2 Doors (ATEDoor)

Doors slide between a closed and open position using velocity interpolation (`VInterpTo`). Key configuration options include:

- `OpenOffset`: the direction and distance the door slides when opened
- `OpenSpeed`: the interpolation speed of the slide movement
- `bStartOpen`: whether the door begins in the open state at level start
- `bAutoClose`: if true, the door automatically closes after `AutoCloseDelay` seconds
- `LinkedButton`: a reference to the button controlling this door; the door locks and unlocks the button in sync with its movement state

4.3 Pressure Plates

Pressure plates remain active only while a character (player or clone) is standing on them. When the character steps off, the plate deactivates and any linked doors close. This mechanic is the primary motivation for the clone — the player cannot be in two places at once, but with a clone holding a plate, they can.

4.4 Movable Floors / Elevators (TEMovablePlatform)

Movable platforms translate between a start position and an end position defined by `EndPositionOffset`. They activate when a character overlaps their trigger box and deactivate (finishing their current movement leg first) when no character is present. Key modes:

- `bLooping = false`: one-way trip from start to end, then stops permanently
- `bLooping = true`: continuously loops between start and end while occupied; rests at the nearest end position when all characters leave
- `bAutoActivate`: starts moving immediately at `BeginPlay` without requiring overlap
- `PauseAtEnds`: a brief pause at each end position before reversing direction

4.5 Laser Hazards (ATELaserHazard)

Laser hazards are directional beams that run from a LaserOriginMesh to a configurable LaserEndComponent (arrow). Every CheckInterval seconds the laser performs a sweep check:

- Clone blocking: the laser first checks all clone actors in the world against three height sample points (feet, mid-body, chest). If any sample is within 80 units of the beam, the laser is flagged as blocked and the player cannot be hit.
- Player hit detection: if not blocked, a sphere sweep along the beam checks for player collision. A hit respawns the player at the nearest active checkpoint.
- On death: the active recording is stopped, the active clone is frozen in place, a camera shake is applied, and a DeathCooldown timer prevents repeated instant deaths.

Lasers visually dim to a dark red when blocked by a clone, providing clear feedback to the player.

4.6 Checkpoints (ATECheckpoint)

Checkpoints are box-trigger actors placed throughout levels. When the player overlaps one, it fires OnCheckpointActivatedGlobal, which TELevelManager listens to in order to update the current respawn location. Checkpoints are sorted by their CheckpointIndex and only advance forward — revisiting an earlier checkpoint does not regress progress. One checkpoint per level must be marked blsStartCheckpoint to serve as the Restart Level destination.

5. HUD & UI (ATEGameHUD)

The HUD is driven by ATEGameHUD, which reads data from the player character and recording component each tick, then exposes that data to a Blueprint widget (WBP_GameHUD) for display. The following information is surfaced:

bIsRecording	Whether a recording is currently in progress
RecordingTimeRemaining	Seconds remaining in the current recording window
RecordingDuration	Total recording window length (used for the progress bar)
bHasRecording	True if recorded frames exist and a clone can be spawned
bCloneActive	Whether a clone is currently alive in the world
bCloneFrozen	Whether the clone has finished playback and is frozen in place
CloneLifetimePercent	Normalised value (0-1) of remaining clone lifetime for the timer bar

6. Technical Architecture

Time Echo uses a modular C++ class architecture built on top of Unreal Engine. Each gameplay system is encapsulated in its own actor or component class, communicating through delegates rather than direct references wherever possible.

6.1 Class Overview

Class	Responsibility
ATimeEchoCharacter	Player character: handles movement, input, and owns the RecordingComponent and ActiveClone reference
TERecordingComponent	Records FTEFrameData at fixed intervals; stores and exposes the recorded frame array
FTEFrameData (struct)	Snapshot of player state at one instant: location, rotation, timestamp, sprint/jump/interact flags
ATECloneActor	Replays recorded frames via interpolation; handles interaction events; broadcasts lifecycle delegates
ATEInteractableBase	Abstract base for all interactable objects; manages linked actor notification (doors)
ATEButton	Concrete interactable; activatable by player E-press or clone proximity interaction
ATEDoor	Sliding door; locks/unlocks its LinkedButton in sync with movement state; supports auto-close
ATELaserHazard	Directional laser beam; checks clone blocking before player hit detection; respawns player on hit
ATEMovablePlatform	Platform that moves between start and end positions; activated by character overlap
ATECheckpoint	Respawn marker; fires global delegate when reached; sorted and tracked by TELevelManager
ATELevelManager	Singleton; discovers and sorts checkpoints; handles ResetToCheckpoint and ResetToStart
ATEGameHUD	Reads player/clone state each tick; drives the Blueprint HUD widget with data bindings

6.2 Key Design Patterns

Delegate-driven communication: all major events (clone spawned, checkpoint reached, door opened, player zapped) use Unreal's DECLARE_DYNAMIC_MULTICAST_DELEGATE macros. This decouples systems and allows Blueprint to bind to C++ events without modifying source.

Frame-rate-independent recording: the recording system captures at fixed time intervals using elapsed time accumulation rather than per-frame capture. Clone playback uses timestamp-based interpolation (FMath::Lerp on location, normalised yaw delta on rotation), ensuring the clone behaves identically regardless of the host machine's frame rate.

Clone collision model: the clone uses the OverlapAll collision profile and disables its CharacterMovementComponent. It is positioned exclusively by the replay system's SetActorLocation calls. This prevents physics interference while still allowing overlap events for platform activation.

Laser blocking algorithm: instead of relying on Unreal's collision to detect clone blocking (which would require matching the laser sweep channel), the laser performs its own proximity check. It samples three height points on each clone's capsule and computes the closest point on the laser segment. If any sample falls within 80 units of the beam, the laser is considered blocked.

7. Level Design

Time Echo contains 20 levels arranged in a carefully sequenced progression. Each level introduces one or more new mechanics, then subsequent levels layer them together. The final two levels serve as a master challenge and a narrative payoff respectively.

7.1 Mechanic Progression Overview

#	Level Name	New Mechanic	Description
1	First Echo	Button & Door	Introduces the button-and-door interaction. The player presses a button to open a door and proceed. Teaches basic interaction.
2	Which One?	Multiple Buttons	Three buttons are present but only one opens the door. The player must experiment to find the correct button, learning that not all interactables do the same thing.
3	Distant Reach	Clone Required	A single button and door are placed far apart — too far for one person to press and pass through simultaneously. The player must record themselves pressing the button, then spawn a clone to hold it open while they walk through.
4	Stand Your Ground	Pressure Plate	Introduces the pressure plate. The door only stays open while something stands on the plate. The player must use their clone to hold the plate while they pass through.
5	Chain Reaction	Multiple Plates (series)	Multiple pressure plates are arranged in sequence; only one is linked to the door. The player must identify the correct plate and use their clone to activate it.
6	Rising Floor	Movable Floor	Introduces a movable floor (elevator). A pressure plate controls the floor height. The player must send their clone to hold the plate while they ride the floor up or down to reach the door.
7	On the Move	Automatic Platform	Introduces an automatic moving platform that activates on overlap. The player must time their movement and their clone's position to ride the platform across a gap.
8	Press in Order	Ordered Buttons	A series of buttons must be pressed in the correct sequence to open the door. The player must plan their recording carefully and execute button presses with proper timing.
9	Floor and Plate	Plates + Moving Floor	Combines pressure plates and movable floors. The player must position their clone on a plate to

			raise or lower a floor section while navigating to the exit.
10	Sequential Plates	Ordered Plates	Similar to Level 8 but using pressure plates instead of buttons. All plates must be activated in the correct order. Requires careful choreography between player and clone.
11	New Room	Room Combination	A separate new room introduces a more complex combination of mechanics seen so far. Increases spatial reasoning requirements with multi-room navigation.
12	Laser Warning	Laser Hazard	Introduces the laser hazard. Touching the laser resets the player to the last checkpoint. The player learns that clones can block the laser beam, allowing them to dash through safely.
13	Blocked Path	Laser + Button	A laser blocks the path to a button. The player must use their clone to block the laser while they press the button to open the door.
14	Shield and Run	Laser + Plate	Combines laser hazards with pressure plates. The player uses their clone as a shield to hold the laser while standing on a pressure plate to progress.
15	Platform Gauntlet	Laser + Platform	A moving platform crosses a laser field. The player must time their clone to block the laser at the right moment while boarding the platform.
16	Three-Way Block	Laser + Multi-mechanic	Multiple lasers and interactables combine. The player must coordinate their clone across different positions to clear a path through the room.
17	Full Circuit	Laser Combination	A more complex laser and puzzle combination requiring the player to use the clone for simultaneous blocking and button activation across the room.
18	Crossfire	Intersecting Lasers	Two lasers are arranged so their beams intersect. The player must position their clone precisely at the intersection point to block both lasers simultaneously, then cross to the exit.
19	The Grand Puzzle	All Mechanics	The final challenge level combines every mechanic introduced throughout the game: buttons, pressure plates, movable floors, automatic platforms, and lasers. Requires mastery of all systems.
20	The Ending	Interactive Finale	A set of pressure plates surrounds a floating button. Stepping on the correct plate lowers the button within reach. The player interacts with the button to trigger the game ending sequence.

7.2 Detailed Level Descriptions

Level 1: First Echo

New Mechanic	Button & Door
Description	Introduces the button-and-door interaction. The player presses a button to open a door and proceed. Teaches basic interaction.

Level 2: Which One?

New Mechanic	Multiple Buttons
Description	Three buttons are present but only one opens the door. The player must experiment to find the correct button, learning that not all interactables do the same thing.

Level 3: Distant Reach

New Mechanic	Clone Required
Description	A single button and door are placed far apart — too far for one person to press and pass through simultaneously. The player must record themselves pressing the button, then spawn a clone to hold it open while they walk through.

Level 4: Stand Your Ground

New Mechanic	Pressure Plate
Description	Introduces the pressure plate. The door only stays open while something stands on the plate. The player must use their clone to hold the plate while they pass through.

Level 5: Chain Reaction

New Mechanic	Multiple Plates (series)
Description	Multiple pressure plates are arranged in sequence; only one is linked to the door. The player must identify the correct plate and use their clone to activate it.

Level 6: Rising Floor

New Mechanic	Movable Floor
Description	Introduces a movable floor (elevator). A pressure plate controls the floor height. The player must send their clone to hold the plate while they ride the floor up or down to reach the door.

Level 7: On the Move

New Mechanic	Automatic Platform
Description	Introduces an automatic moving platform that activates on overlap. The player must time their movement and their clone's position to ride the platform across a gap.

Level 8: Press in Order

New Mechanic	Ordered Buttons
Description	A series of buttons must be pressed in the correct sequence to open the door. The player must plan their recording carefully and execute button presses with proper timing.

Level 9: Floor and Plate

New Mechanic	Plates + Moving Floor
Description	Combines pressure plates and movable floors. The player must position their clone on a plate to raise or lower a floor section while navigating to the exit.

Level 10: Sequential Plates

New Mechanic	Ordered Plates
Description	Similar to Level 8 but using pressure plates instead of buttons. All plates must be activated in the correct order. Requires careful choreography between player and clone.

Level 11: New Room

New Mechanic	Room Combination
Description	A separate new room introduces a more complex combination of mechanics seen so far. Increases spatial reasoning requirements with multi-room navigation.

Level 12: Laser Warning

New Mechanic	Laser Hazard
Description	Introduces the laser hazard. Touching the laser resets the player to the last checkpoint. The player learns that clones can block the laser beam, allowing them to dash through safely.

Level 13: Blocked Path

New Mechanic	Laser + Button
Description	A laser blocks the path to a button. The player must use their clone to block the laser while they press the button to open the door.

Level 14: Shield and Run

New Mechanic	Laser + Plate
Description	Combines laser hazards with pressure plates. The player uses their clone as a shield to hold the laser while standing on a pressure plate to progress.

Level 15: Platform Gauntlet

New Mechanic	Laser + Platform
Description	A moving platform crosses a laser field. The player must time their clone to block the laser at the right moment while boarding the platform.

Level 16: Three-Way Block

New Mechanic	Laser + Multi-mechanic
Description	Multiple lasers and interactables combine. The player must coordinate their clone across different positions to clear a path through the room.

Level 17: Full Circuit

New Mechanic	Laser Combination
Description	A more complex laser and puzzle combination requiring the player to use the clone for simultaneous blocking and button activation across the room.

Level 18: Crossfire

New Mechanic	Intersecting Lasers
Description	Two lasers are arranged so their beams intersect. The player must position their clone precisely at the intersection point to block both lasers simultaneously, then cross to the exit.

Level 19: The Grand Puzzle

New Mechanic	All Mechanics
Description	The final challenge level combines every mechanic introduced throughout the game: buttons, pressure plates, movable floors, automatic platforms, and lasers. Requires mastery of all systems.

Level 20: The Ending

New Mechanic	Interactive Finale
Description	A set of pressure plates surrounds a floating button. Stepping on the correct plate lowers the button within reach. The player interacts with the button to trigger the game ending sequence.

8. Future Extensions

The Time Echo system is designed to be modular and scalable. Potential extensions include:

- Multiple simultaneous clones: allow the player to spawn several clones from separate recordings, enabling more complex multi-actor puzzle solutions
- Longer recording durations: extend the 10-second cap to support larger puzzle spaces
- Multi-stage puzzles: levels where completing one phase unlocks a new area within the same map
- Advanced environmental interactions: conveyor belts, rotating hazards, timed switches
- Time manipulation mechanics: rewind to an earlier point in a recording, or create branching clone paths
- New hazard types: timed traps, proximity sensors, or moving laser emitters
- Narrative layer: environmental storytelling elements triggered by reaching checkpoints